

ActInf Livestream #014.2 ~ The Math is not the Territory: Navigating the Free Energy Principle

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hello and welcome to the active inference lab this is the active inference live stream today we are in active inference live stream 14.2 on january 26 2021 today is going to be a great conversation so thanks everyone who's here live and thanks everybody who's watching live and in replay welcome to the active inference lab the active inference lab as i share my screen is an experiment in online team communication learning and practice related to active inference you can find us on our website at twitter gmail youtube or our public key based user name and uh or public key based team and shared username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and as far as video etiquette for live stream goes check out the checklist and remember to fill out the survey at the end to give us feedback and then during the conversation we'll mute and we'll raise our hands so we can hear from everybody to learn more about active inference live streams for 2021 you can go to this [rb.gui](#) link where you'll find a spreadsheet today we are on january 26 14.2 with our second participatory group discussion on the math is not the territory with author mel andrews and next week as we head into february we'll go to a different paper so check out this spreadsheet if you want to read the papers ahead of time and get familiarized with them today in active infant stream 14.2 we're going to go through our introductions and warm-ups and then we're going to walk through 14.2 again being a follow-up on the math is not the territory navigating the free energy principle by mel andrews in october 2020 and we're able just to kind of chill talk raise some questions that come to mind as we continue and then next week we're going to be on a different wavelength always welcome to have participation so introductions and warm-ups it's a nice group of seven today we can introduce ourselves and feel free to give any kind of introduction or check-in you'd like and then we'll pass it to somebody who hasn't spoken so i'm daniel i'm in california and i will pass it to yvonne hello my name is ivan i'm from moscow russia and i pass it to marco let's go to alex first [Music] hi everyone i'm alex i'm also in

moscow russia and i'm a researcher in systems management school and i pass it to stephen
hello i'm steven i'm based in toronto canada i'm doing a practice based phd at canterbury christ
church university and i'm going to pass it over to cherry uh sarah if you want to otherwise let's
go to marco hi i'm marco uh based in holland and i'm i'm just a active enthusiast i'll pass it to
mel hi there i'm mel andrews i'm doing a phd in adjacent stuff at the university of cincinnati is
that it is not all of us yes cool and then sarah any time you want to jump in is chill so today oh
yeah go ahead sarah uh hi i'm sarah i'm a person interested in stuff and things mel's talk is
awesome here i am what else can be said so for the for the warm-up questions we'll just dwell
on them they're just openings and just opportunities for us to raise the kinds of things we've
been thinking about today and so what is the uh what's something you're excited about today
and then also what did the article or last week's discussion make you curious about so
something you're bringing exciting today or just a question or even if it's adjacent or it's not a
quote from the paper or anything like that like who are we that has brought us today to this
conversation marco and then i believe it's stephen but stephen and mel if you could add your
name on jitsi but marco and then uh stephen yeah um well last last week we uh ended the with
uh please three pill now me and uh alex keeper on the uh uh the prospect of continuing
discussion about the relation between a thermodynamic entropy and information so very much
excited to explore that because i think that was very crucial parts in the paper that deserve
more on cool steven and then anyone else yeah i'm i'm bringing it an interesting um
dimensionality and and how that relates in like high dimensional spaces low dimension spaces
and entropy so that that was something that this paper really kind of got me thinking about so i
think while it seems very abstract okay could you also speak a little bit more directly into your
microphone um but thanks for that so yes yes okay so definitely um yes philosophy paper
abstract general theoretical a lot of adjectives that we hear with philosophy but then this is
almost about the philosophy of utility and about the reality of the models that we make so it's
kind of uh has a hand in both of those worlds which is what makes it an interesting thing to
think about i'm also curious about that intersection of how do we generalize and abstract
correctly in order that we can make more effective action and i'll just put up the last two
questions and anyone else is welcome to raise their hand what is something that you liked or
remembered about the paper and then what is something that you're wondering about or would
like to have resolved by the end of the um paper or by the end of the discussion today so one
thing i'll just start with is uh if the math and yvonne you can you too please thank you so if the
math is not the territory then what what is our metaphor here like what is the territory or what

is the math if we're gonna go with the title of the paper that the math isn't the territory what is our uh complete spatial metaphor if that's even helpful to walk down but it's just um uh yeah just a few thoughts anyone else want to raise anything i i guess i have a maybe a general question which is um for mel like uh almost like a one to ten or how how mel maybe feels that fep contributes to like epistemic unfolding like the best possible tools to allow us to ask the most relevant questions that's yeah mel yeah i think i so i i think i i am came at this with maybe a kind of naive and outdated view of of the scientific method and i really didn't like very general very abstract models and i really didn't like use of metaphor um because i saw a lot of the the kind of mistakes that were being made with very abstract models and with with kind of analogy okay please do not yeah thanks make sure that everyone's audio let's take a few minutes before the hour begins so that we're here on time so that everything is prepared but thanks a lot the audience also appreciates it now continue what happened oh there's just audio on a loop cool not me knowers yes not you although i was also late yeah so i i i tended to think of um certain uh approaches in sciences as being kind of bad because i saw a lot of of mistakes that were being made with them so a lot of um kind of reification where we start with an analogy like oh you know we've got selfish genes or oh we've got you know borrowing models from physics into biology you know and then and then um confusing them refining them um but the the mistakes are not are not the methods right there's a distinction between the mistakes that come along with with certain methods and methods themselves and there are mistakes that come along with everything and um without with any kind of approach you're going to take complaints and um especially when it comes to sort of uncharted waters um you really need to be more creative um you really need more generic models you need to be borrowing models from one domain and importing it into the new domain kind of casting in that right so when when you've got um you know very simple systems and in like classical physics and chemistry um you can kind of build up direct models um without doing much analogizing right um but when it comes to dealing with something like the human brain or an ecosystem things you know or even like social structures were um where we're just sort of out of our gap you need a lot more looseness and for that reason i think when it comes to the kind of systems that we want to understand in future science things like the fcp are actually going to be the way to go cool well this whole future science i'm sure that's something we can return to but this was a really great point that when we're exploring or constructing a new area we do need to borrow ideas vocabulary methods etc because otherwise what is there to be discussing or converging on but these come with baggage preconception and uh just incompleteness

because they're not the godelian incompleteness just they're metaphors they're not total one-to-one of everything so how do we navigate borrowing from the past but also moving into something that has different attributes so stephen then anyone else that's a very good point in terms of um you mentioned about this like does it belong to science so we often think about is something in engineering or applied science or is it science but then i suppose there's a question of when is something useful to have in a philosophical sense somewhere in between science and philosophy so not completely philosophy but maybe there is another category of um foundational knowing um that we kind of then you then use to build other things upon so maybe there's something in that yeah it's it's almost like this question about whether the natural kinds are disjoint or not and then okay all the stuff or all the approaches that we can take there's science and not science is that a clean break or is there multiple kinds of non-science even multiple kinds of science so this is the demarcation problem and maybe the simplest demarcation would just be uh the bright line and line in the sand but what if the demarcation problem is a little bit more nuanced and it has islands or it has other attributes that are not just a line across the gymnasium floor so mel i'd be curious how did you think about the demarcation problem early on in your research or when you initiated this and then where has that taken you on the demarcation problem which is again about what is science and what isn't science yeah i'm um unfortunately i think the longer i spend um my background isn't cognitive science i was doing i i had some bio background um and some psychic concerts um so so this is the first time i've been you know full-time a philosopher um and i find that the longer that i immerse myself in philosophy um the more equiney and i become which is it sort of to my chagrin i i didn't um expect to see myself going that direction what is that sorry what does that mean the quiney in in this context yeah um naturalism which means no backdoor to knowledge the um the only access we have to knowledge is empirical so uh there's there's like science and there's commentary on science um and and there's this i view the relationship as as between science and philosophy um as being kind of intrinsically connected they need they need each other you know the philosophy comes up with new questions and that they don't know how to ask yet empirically and then we figure out over time ways to go about these these questions empirically and we we come up with methodologies and then the methodologies fail they hit a they hit a standstill and then um philosophers have to come in and say well look you know you you've kind of arbitrarily narrowed yourself to the outset we need to start thinking a different way right um and i don't think i think like you know in contrast to what what sort of quan there's a there's a kind of popular um thought that goes

along with naturalism nowadays which is just that philosophy is this kind of finite stock of questions and as we hand off as we as we pull the questions out of magic hat and have them off between empirical sciences we run out of questions and so you know fortunately in the near future philosophers will run out of questions to ask because the scientists will have solved all the questions and then you know then we'll be out of block right which i think is just kind of hopelessly ridiculous i mean it's it's really a it's a really simplistic picture and i'm you know fortunately for me i don't think will ever be out of a job i don't think scientists will ever be out of a job i don't think philosophers will ever be out of a job i think i think there will always be questions for as long as we persist um but uh i i do think that ultimately i am i am starting to come around to the idea that there's just an empirical route to knowledge and um that means to me where i'm at with that right now in in kind of processing that and digesting that it's um that there may not be a sharp dividing like there might not be a a bright line distinction between science and not fights um and what i really hate about this is is i i really want i really want um math to be entirely conventional i want math to be invented not discovered constructed and to be a naturalist about mathematics you have to kind of record it more reality and naturalness than i i want to but you know so that's where i'm at that's my i might have to but i might have to bite that bullet um what what uh stephen was saying um there's a so the the two people what you were saying about kind of like foundational things that come before um this ties into what uh is being said by uh inez de politico and um what was said by um matteo colombo and cory wright in a 2017 piece um about about first principles um so there's a there's a piece 2017 piece called first principles in the life sciences something in mechanism um i think it's in bjps why matteo colombo and quarter right about first principles and the fep it's about the fpp but but but this first principles approach in the live chat put it in there so cool people can check that out do you want to continue yeah or sarah do you want to uh go ahead after you've adjusted your chair but go ahead yeah i'm pulling a wire from under my desk um yeah i have a question for you mel um i'm i'm as a person a new student in philosophy of science i'm bumping and somebody who came from physics i'm i'm bumping up against this like dynamic that i see between philosophers and science where you know science will come up with something and then there's this kind of like ankle-biter dynamic where like philosophers will be like well what about yeah but what about and i'm just i mean i want a more positive conception of what philosophy can be and the only one so far that i see that's really clear and i'm a new student so it's not like so i'm probably missing a lot but like i really like chris fields because he's he's more like not afraid to be sci-fi you know like he's not afraid to have a

speculation which is originally what a lot of philosophers did so i'm like i just i really want advice as like a student about what what how you feel that dynamic can be we'll go mel with a direct ask and then marco then stephen yeah chris hughes is awesome he gave a talk recently to our um to one of our working groups and that was fascinating a lot of it was over my head it's just introducing so many new concepts but really fantastic and he's joining in march early march for active17 he'll be on yes what are we reading tbd to be determined but we'll find out very soon fantastic yes awesome yeah and there's something i read recently i'm there's a long story um i i wouldn't i don't want anyone thinking that i would just go out and read heidegger because i wouldn't um what happened was there's a long story about tracking down quine's typewriter and and like writing back and forth with quen's son and finding a heidegger scholar who had just acquired the typewriter of martin heidegger who i used to correspond with as a kid um but i ended up reading some height in in part of as a sort of part of writing a piece up with a heidegger scholar um and uh just just a couple paragraphs but but there's um there was this there's a line about um that uh there's this sort of um and this is you know this isn't like at the time of like the vienna historical at the time of sort of the the positivism right and and and there was this line about logic and philosophy of science kind of like limping after the sciences and sweeping things up and putting cobbling things back together you know like cleaning up after science in this very um kind of horrible lame way um and and it wasn't contributing anything of its own it was sort of doing like you know like logical reconstruction of scientific developments but but nothing that actually really you know contributed to to furthering the ends of science and um haiduker was was declining that and saying that we need uh a generative logic that we need to go back to the model of um the relationship between philosophy and science um of the time of like plato and uh aristotle and and the you know thinking ahead of science and i i i just think i i very much think that's what we need and i think to some extent um there's still very much like a live tradition in philosophy that is stuck in this kind of um we're just we're just describing we're just doing the bookkeeping right um but at the same time there's i think very much a live and active tradition of philosophy of science today that is that is really thinking ahead of science and um coming up with new questions problematizing things in a way that um that leads to real breakthroughs cool marco then steve thanks yeah i really like the discussion so far um i i i i would like to point to the uh notions of uh abductive inference and the uh exploration expectation um um complementarity uh to maybe approach this relation of science and philosophy so i mean i i also have my qualms with philosophy uh like sarah and also noted i i personally really hate the um excessive well in

my in my view a test of trend to uh seem authoritative and absolute tears and these weird fetishes enough of universal truths and stuff i i i'm not fun of that what i'm more fond of is the it's almost for me it's an art right so it's the art of taking something a given sets of assertions or propositions whatever and then generating more territory to explore from that right so the way i see it is for under active inference knowledge is generative right and so that generativity can be seen as leading to abductive inference so it generates new hypotheses um but it can also be seen as imagination and um the that level that that that step of generativity is then itself rooted in what's been generated upon a direct contact with empirical reality and so for me it's like an intermediate kind of step it's like a a detour you create all these hypotheses you explore the maps you create it yourself and then you you can return to the imperial grounding and so that's kind of how i see uh philosophy it's kind of a deep work and like mel said they're they're inextricably connected without an empirical basis you have no grounds uh on which to imagine and explore and reconceive right but at the same time you need to pass back everything that's recons reconceptualized explored imagined these philosophical uh modes of inquiry can then return to the scientific realm so it's nice back and forth at least that's that's how i see it and then and like mel said there's no clear demarcation yeah i don't also don't think there's a clear demarcation it's simply for me the gradient of how well grounded it is in the empirical uh roots of these inferential processes how much weight does it have um where the other extreme of the spectrum would be that you generate a lot of understanding or sense or reduced uncertainty by virtue of this reflexive engagement with your knowledge i think ryan smith had a paper on model uh expansion and elaboration so maybe that's also a nice connection that makes sense cool but yes um you use them sorry say again who that was that had an idea you just right at the end oh right yeah yeah i think ryan smith has something on model expansion and stuff um but i'm not sure um but also you can also put the paper on abductive inference and curiosity and insight which is actually the paper i co-authored with the pearl in 2017 um yeah but that's kind of what i really always like abductive influence it's well yeah inductive reasoning is when you go from the particular to the general and then deductive is when you go from the general or the philosophical idea to the specific and then abductive is uh when you go sort of laterally like your your abductor or something like that it's off to the side and so that's this imaginative or generative way that we're talking about as kind of threading between science with the exploit and philosophy with the explorer are we ever going to be trapped into this dialectic philosophy's broadening our horizon and science is narrowing them and then it's almost like there's this abductive sideway

that we can take where there's a fruitful discussion and so that's kind of the uh niche and the intersection we're reimagining so stephen and then anyone else who raises their hand stephen then mel yeah i think that the actually that's a very good point about abduction and in a way you have to say everything that we know started with abduction from a child because it's only with time that you start to set um i i often think with induction is you've got a target and you try and narrow the gap you know so you but you've got a target so where's the target come from how did you know what the target was um and um you've also got that question then of so i think this is these these questions about how they get applied are very very relevant to um to bringing in this question of you know are you applying it to help scientists or are you applying the philosophy to help you know you decide to sit and how am i going to sit in church or how am i going to spiritually be in myself so you've got this kind of and i think this is where you're starting to see that the same philosophical questions being addressed with the same framework as opposed to it being like okay that's for them you know it's starting to break down some of the silos i think of how it hits the ground and i think that's quite important yeah now yeah it's sort of sort of what mark grove said but also sort of something you said in the beginning which is um i guess dan you said this um which is if the map is not the territory what isn't that right if the if the math is not deterred for what is in that um uh and i think there's there's a kind of um classical position in in i guess both science um and maybe even also the the popular conception of science um which is very prevalent in modern times which is that there are there are like a finite number of atoms in the universe and you know we're we're picking away at um psychology and we're picking away at the social sciences and we're picking away at biology and chemistry and and so on and so forth but there is some some fundamental level of reality you know whether that's atoms or or you know whether that's the quantum level or whether that's string theory or what have you but there's there's some ultimate level of reality and eventually you know we're going to figure out how the psychology reduces to the biology how the social reduces to the psychology how the biology reduces to the chemistry and fitness right and then and then we'll be done and i think that this is this is a very um mistaken outdated view of what um what reality is like but more more so what are our kind of epistemic acquaintance with reality it's like i think it's it is as as mark was saying far more far more degenerative right where we're um we're instead of instead of uh instead kind of constantly reducing the the allowable frames of reference for for understanding the world we're actually um constantly generating them you know we we are we should be doing right something like model selection some models some theories some accounts and the

sciences do similar work to others and are not as good and those get weeded out right but we're also constantly coming up with new approaches because any frame of reference right any any model any theory that we take is not purely a reflection of the world out there but will also be a reflection of our kind of epistemic aims will also be a reflection of of what we want to get out of out of that way of relating to the world and so there there you can see how there's gonna be you know a kind of infinity of of possible ways of getting at the world because they all reflect our are kind of ultimately subjective aims as researchers yes very nice point about the reductionism is misguided just the fantasy we're going to keep on reducing to smaller and smaller pieces so let's think about the map if someone says i want a map of california and someone goes oh no no that's already been solved it's made up of dirt see no i don't want that i want to be on a road trip or i want to be visiting my friends so when we reintroduce action and pragmatism and utility the argument against reductionism is very obvious which is okay it may be that as a substrate but that's not actually how we're moving forward and then it's interesting that there's a computer science algorithm called map reduce now it's a different map in a different reduce but it's almost like action and inference we map we do sense making and then we reduce the scope so we map and we we sketch out in very broad strokes and then we have to go into that reductionist mode to go into a little bit more detail about what is actually on the road or in the ocean ahead but we're mapping and reducing or acting and inferring with an eye towards the direction that we're already moving in not just because we're gonna uh get to the bottom of action or the bottom of inference and then just call it a day like just like you said it's an infinite game we're gonna keep on coming up with questions and keep on coming up with measurements for the dialogue between scientists and philosophers and ultimately just participants in the knowledge enterprise whatever somebody's title or profession happens to be so marco than anyone else thanks uh daniel that was very beautiful it's it's the image evoked for me is like uh a directed cycles of contraction and expansion which uh i believe is also very active in french but anyways um so so um about the uh question about the maths and language and everything so i my improvised pitch is the maths or the mathematical model of active influence effectively describes how the terrain maps itself so if terrain is the thing itself or the actual reality of the brain or any ecosystem then active uniform is basically saying well all the constituents of this ecosystem or brain for example is constantly deferring each other it's constantly engaging in mutual and reflexive inference and all these inferences will then lead to models of each other and effectively they all do this kind of subjective mapping right but all these subjective maps over skills and levels then also need

to calibrate to each other and so yeah if we have to play with words i would say the maths describe how the terrain maps itself which i think is you know quite nice um also uh it's very strongly echo uh what mel said i was very briefly worried that that she would actually argue for reductionism i should know better of course um yes so so if it's appreciated um i really like thinking in terms of uh trees so there's a book by uh julio africa sorry but my favorite phrase is um trees of the brain roots of the mind uh for someone some reason that always stuck with me and and basically it's because there's these generative trees lead to roots at yet another level and um again kind to a uh the previous discussion uh so the reduction in story is more about you know sort of essentialism properties and objects and kinds which are not a phenomenon um but complementing that kind of uh is perceptualism and so i really again think that this generative perspective is in spirit with perceptualism is how how do these processes actually structuralize themselves right how do these flows and dynamics attain some kind of regular structure and then that regular structure will be in the form of flows and generative relations um so it's like generative scaffolds and i think that would be an interesting angle to approach anthology as well cool oh oh sorry and one more one more points if you can is i also want to echo what mel said about pluralism right so i think looking at active infants in my opinion uh i pray enough to talk about knowledge because then you get the baggage of correspondence uh notions of truth and i i think that's distracting if you just look at it from an active inferential perspective actually all knowledge the value of knowledge is going to be contextual it's going to be contingent upon the actual agents right even though we can say oh it's just the accuracy of the states in the world as as believer as mass but these states don't have to conform to some objective truth it's more about given the states that it has models if it acts based on these states how accurate would it be and then there's a bit of a i guess a pragmatist account of truth maybe i don't know um but i guess what i'm saying is that that kind of implies also for epistemology that it's about the normative contextuality and that we really should pursue um pluralism in science which yeah cool thanks marco we'll go dave then mel so i can't hear you dave but you're unmuted so maybe try to reload or i'm not sure what's going on there but mel go ahead or do you hear dave i don't hear him yeah i don't okay so yeah mel go for it but i know that you can hear us steve but we're on a live stream so we can't figure it out at this moment so mel go for it um i uh yeah i just want to say something effective um i think that that what i guess we're all sort of converging to describe that um it fits in very very well with the kind of bayesian view at least at least uh the bayesian view of science in the future there are some kind of study old bayesian views of of scientific updating um of the late 20th century mid late 20th

century that um i think we've we've surpassed but but i think if we view um scientists as kind of subjective bayesian agents coming up with like different kind of parallel models of the world and adding evidence to these and kind of retiring models that don't that don't um that aren't useful in a way um i think this is this is very much what we're describing there yep it's about these metaphors and we're talking about understanding the relationship of science and non-science that's the demarcation project and we've heard a couple of different metaphors in this introductory section maybe dave if you want to reload and try again but in this introductory section we talked about one was like a spectator model with the scientists doing the work and then there's the peanut gallery and the philosophers are either cleaning up behind them or are confusing in their trail and that was the following after science metaphor and we talked about how that was inadequate because there's actually a really fruitful bi-directional generative relationship and that reminds me a little bit of a live stream so it's almost like in the context of uh on the live stream the people who are on the panel are the scientists the observable states being emitted being viewable to the world the empirical states are like the state of the live stream and then we have all of these curious philosophers whatever their title or position is in the live chat and they're commenting asking questions and by doing so engaging in this real time dialogue so we actually have this instantaneous system where we can talk about how the observable states are in feedback with what people are saying at that moment so it's sort of like again it's just a metaphor it's not that it carries every single dimension of the metaphor through but it's just really interesting that there's a discussion that can be held amongst many people in the live chat that's kind of like the philosophy literature and then the people who are in the live stream could just not be looking at the live chat and that would be like a scientist who just tunes out and chooses not to engage with that discourse or the people on the live stream might be obsessed with what is happening in the live chat and that could actually distract them and help them or prevent them from carrying out the aims of the stream so it's a pretty fun way that our new technological affordances and the rapidity of them are helping us understand the relationships and the bi-directional dialogue between those who have for example technical expertise and those who are just learning like we've seen on these streams and in the model streams so dave let's see if we can hear you and then uh yep how's that yep go for it and then sarah okay yeah i'm glad you uh mentioned a road trip that um reminds me of my puzzlement why people are talking about homeostasis rather than allostasis it seems we've got the you know and this is a puzzle going back to freud's uh proposal for a sign project for a scientific psychology we want to get to nirvana we want to

stop everything we want to be perfectly happy in the eternal now but then something happens and it's horrible uh that's preposterous that's not how any organism ever acts organisms want to go through a series of states we want to be tired so that we can rest and we want to rest so we can run around and get tired and be a little hungry so that we can enjoy the food more we want to follow trajectories we want to take road trips um was the notion of allostasis proven to be useless or to just become unpopular because homeostasis seems to fit with somebody's mathematical formalisms i'll give a quick response on that and then we'll go sarah marco so homeostasis is like the first pass absolute value you say your temperature returns to a certain number so it's homeostatic it's thermostatic you know it's just like a thermostat and so at the first pass somebody who's looking at just the absolute value of temperature or a blood sugar might note that it returns to a certain point and then in a static conception of the universe rather than this process driven understanding it's like right if it's returning to this value of 10 then it's getting homeostatically attracted back to 10. but you're taking a look at the process and the derivatives the rates of change and you're like wait it never stops in fact it couldn't and it shouldn't stop so why don't we just focus on the fact that it's doing this uh iterate road trip that's yes it's returning to 10 a bunch of times but shouldn't we highlight the fact that it's on a circular road trip rather than just say yeah it's at 10. so it's really uh two sides the same coin whether one chooses to highlight the instantaneous empirical observation or highlight the process and the approach that the system is actually taking to result in those observations so just like many other things we're looking beyond the superficial to the generative and to the underlying models and that's what is revealing all of these patterns across systems so sarah and then marco um i'm i'm stuck on something math-wise uh but i think it you know you could describe it as conceptual as well um you know when i look at uh as far as i've gotten anyway with active instruments and fep you know it's all predicated on bayesianism and bayesianism seems to have a kind of helios in the sense that you do priors and you go forward but um uh i don't know i kind of bump up against like well wait a minute but what about like you know built into fvp there's like a feedback type mechanism but it's kind of built on a substrate that is going forward and i find that kind of interesting like i wonder about the limitations around um and i'm not uh well enough diverse to to like say anything more specific but another little side data point um which hopefully isn't a distraction from the question is you know you see a lot now in in descriptions of complexity they'll talk about well you need something that's of high dimensionality and that's that description is very much predicated on the methods that we use to to to get those dimensions you know it's predicated on the math um techniques and i

i get uh even when i think about this high dimensionality thing i keep coming back to this like well yeah but you've got this like you know f as a function of x or x, y, z you know like it's not there's no feedback in the in the actual heuristic well no there isn't a heuristic in in the end result or in the in the end um and you know like fep like let's say it converges it doesn't quite converge like markup blankets to or like the chain mcmc does but i don't know you get to something that's close by energy minimization but in the end you get to a thing where you're like where you like something is a function of da da da and it never goes back around it okay that's just i hope that's a useful question if anybody has any thoughts around it marco and then if anyone wants to raise their hand to speak to that yeah that was very interesting um uh wait oh yeah well daniel already did most of it but i also want to note that um it's also really what sarah's talking about so there's this issue of um having too much freedom in how you characterize the system right it's it's not really um well defined how you're going to define a state space especially when we're talking about a complex set of systems that model the environment so then we have to not talk about simply physical observables but but more the structures that they constitute so for example with homeostasis and alastairs it's like daniel said there's two sides of the same coin and it needs to be in reference to something else so you can say homeostasis in terms of one perfect temperature and just try to stay there and then add to that uh next to it the processes of anticipation which then gets you stasis but another way to also look at it is for example you can also say instead of having one perfect state or a value and also talk about viable sets for example the domain or boundary in which you in which the agents will survive or remain viable and so you can say okay then that is the state being in the boundary being in a viable sense then it reduces to one point again um which again relates to i think what what sarah's talking about i think um was a lot i think very interesting um so about the function thing right so so for example you can have a a a of blankets it's enacting certain functions and it's not built in the loop the loop is not built into the function but the function enacts a process right it generates perturbation activities flows whatever that expresses into the environment with the expectation that it expresses itself back to them and so that's left implicit i'm not sure if that makes sense also you said something about um things moving forward and i'm not sure what you meant about it but it did feel like it a very interesting angle you were pursuing so if you could elaborate it out i mean it was it was just really simply like it was just you know you have this idea of prior so there's a kind of a teleos you're like going forward in the way you minimize energy or whatever oh so well personally i feel like the metaphor that then is the problem right it's not necessarily forward it's it's optimizing or calibrating or

tuning um but i think maybe i'm just still an understanding um but but it's a constant yeah you're that's just that wasn't the right i agree yeah the moving forward is um related to historiosity and to developmental thinking so understanding although there might be attributes like body temperature that are returned to so that's the homeostasis side of the coin in another sense there is no state that the organism returns to so it's about how will we understand as you pointed the teleos the end well there's this local end directedness as the prior is updated but then we're almost using this short-ended telios because we don't know the ultimate end game and as you point out it doesn't necessarily have to converge it just has to be the most effective model in that ensemble in that niche at that time and that's who does well on that statistics test or in that you know ocean ecosystem so that's sort of this updating model and um yeah mel and then i'll say something about the dimensionality yeah and i think i think the active inference 300 principle kind of framework tends to emphasize i don't i don't know that this is speaking exactly to the questions that you're going through but um there's a there's a way in which it it kind of looks to to kind of hierarchical architecture at the exclusion of of parallel architecture um we don't we we don't often talk about like the jockeying of beliefs at the same level right is there any interaction between kind of like models that exist in in parallel that kind of thing um and i think that there are plenty of you know cognitive phenomena that get mixed that way or or um in in biology right in in organismal biology you know we miss stuff because um the fep doesn't um kind of readily accommodate that as i as i have come to understand it and use it um i don't think it's very good at handling that and so that's that's one way in which you're sort of constraining uh how highly conceived of kind of um epidemic agents engaging with the world um and and there seems to be a sort of i guess i guess you could say that there seems to be a directionality to it that is um maybe misleading that's that's the time hypercube fep question with like the gradient descent and the moving forward through time what if it's kind of ironically circular and that we're experiencing it in the downhill direction but the hill is just there and we're moving through it experientially in a way that is prospective and it becomes pathological and not adaptive when it's not prospective that would be like prioritizing retrospection without an eye towards anticipation and those systems fail to exist so we end up having this like distilled and extracted understanding of our past only partial aspects of it with an action-oriented view towards an uncertain future so that's why fep is such an interesting framework and as mel just pointed out it's not like it's a lateral alternative to how the mitochondria works or how a protein folds it's not really a scientific idea at that level but again to take it to this initial claim of the paper the claim is that the fep can be taken

to belong properly to the domain of so then we're talking about what is in the domain of science if we're going to be in science landia is there only one kind of model are there multiple kinds of models are those models hierarchical are they hierarchical how do they depend on one another who is involved in producing and communicating about those models these are some of the very interesting questions that are almost within the demarcation problem once you're on one side the line or the other there's still more questions again highlighting the fact that it's going to be an infinite game with researchers and philosophers so marco and then anyone else yeah i apologize if not actually covered this i couldn't hear it properly but um i just wanted to add that at least in my view the idea of tails for me is at least i don't like it because i i have an allergy for everything that reeks of essentialism so telos kind of feels like something universal universalistic it's almost absolutely great it's like what is the kids of this thing right but that breaks down when we're talking about entities or systems that are exactly defined in terms of their context right so so each of these systems embedded in a big mutually inferring ecosystem they are at the mercy of their context they don't see the world they only see the larger worlds through the directly adjacent systems with which they stand in an inferential relationship with right so for me the enormous principle of fep is the theological principle as in like like i dan also notes this that the t loss kind of is constantly changing but there's nothing else to say nothing yeah it's not different than saying that there is a normative principle that is defined in terms of the relation of the system and its context and so that's what i find beautiful right so one of my key phrases is contextuality um because what's effectively happening with these systems is they they're they're they're enacting inferential context for me maybe nice parallel is because because daniel knows that the livestream parallel is a domain of discourse so what we're constantly doing in a live stream is we're cultivating or constructing and acting upon a domain of discourse we keep adding new elements to it new perspective new frames and then domain of discourse expands but it's the same thing for these inferential systems these marketing magnets right they are constantly changing their domain of discourse when they reconnect with other inferential adjacencies or let's say interlocutors they change their domain of discourse because that domain is defined by the adjacent systems at least in my opinion this is again a bit speculative away from the brown's truth but i'm curious to hear what you think the patterns of discourse define the interface so it's not just that there's the essentialist nodes that are then like bumper cars on a you know in a chat room who are engaging in this thing called discourse it's like it's a co-construction just like you pointed out with the relational and the contextual or the ecological teleology the purpose is multi-scale and

it's it's relational and then also teleology is just one of the quadrants in the two by two with uh aristotle's four wise the tin bergens four wise so there's the teleological cause but also there are the other quadrants so there's other aspects of why that we might be interested in and at any given moment the why that's interesting or meaningful um is itself situational but there's also a pluralism there so it's kind of like we're weaving these similar ideas of relational thinking pluralistic thinking and then we're finding that we're kind of pluralistic across issues like we're okay with people having different perspectives in a live stream we want to have a plurality in the literature but then at the same time that's that expand cycle and then there has to be the reduce cycle because if we have pluralism that leads to somebody thinking that we should capsize our ship it's not that it's uh you have to do it on all possible universes this is like simply wrong it might be enough for someone just to say it's not my preference it's not my vector that this ship get capsized so that's not a policy i want to engage in so it's kind of like this moderated and this uh pragmatically oriented dialogue with philosophy and science stephen and then anyone else and that's this pragmatism and being able to move between scales the the idea and it talks in the paper you know about thingness the idea of the something being a thing even if it's not a living thing you've got that and i'm curious it didn't actually quite doesn't actually quote the the markovian monism sort of tag in there as that so i'll be curious if that that was a particular choice they didn't get locked in by that um but um this this idea of dimensionality is if the free energy principle or is based on entropic change and entropic is is dimensionless to some extent it's just a scalable eigenvector sort of um i don't know what you'd even call it quality and then it goes into thing this then it goes into some sort of process theory which has to have dimensions at some point you know so you know we want you to in some ways once you go into active inference and there has to be dynamics across things you start to have dimensions or maybe certainly once you have something that's nested inside something else and that's that um i just think that's really interesting i'll be curious what mel what male's thoughts are about how the drops in and out between thingness and say pure entropy and um and and just maybe why she didn't put in markovian monism was that a choice for um sort of reasons of not getting caught up in it or does she not agree with it or something like that could you just briefly introduce what markovian modernism is or why is it exciting or relevant to you okay so well markovian monisms um is the kind of overarching overarching philosophical um uh sort of uh paradigm where the um the the the idea of something being in the world is because there's there is a differentiation between what's outside and what's inside and therefore there's a sense and and the nature of how that inside and outside dynamic is

structured even be it a stone in terms of the stones stays as being the stones that somehow the forces inside balance its relationship to the environment outside and therefore there isn't um there isn't this dualism the stone and purely outside the stone as a complete separation there's this somehow um emerging so everything is to my understanding this is is monism is kind of a an underlying kind of unifying kind of uh dynamic that we um which sort of builds on sort of ideas of david bone and the kind of um that those sort of principles that there's something in the background which is um more um implicit in our existence if i've said it right but that's my understanding okay it's definitely a uh you know big and deserving topic usually the monism would mean that there's basically one kind of thing dualism would be the idea that there's perhaps two kinds of things so the one type of thing in monism whether it's matter or whether it's mind is being manifested differentially these two views often converge because the monus will say yes but it looks like two things or the duels will say yes but they're interconvertible or they associate with each other so it's a little bit of a weedy area so if there's any thoughts on that where you can happily hear that otherwise let's um maybe slowly walk some of these slides and just highlight some of the very nice points of the work itself and then just think about how that kind of springboards us into 2021 and beyond why not sound good yep all right cool yeah go first and then we'll continue on i could yeah i could respond quickly um so there there's a paper um a few years ago that presented um one view on consciousness that that carl christian was a uh an author on um and i think the the phrase that stuck was something about the cartesian theater so it was it was very much a um kind of classical dualist uh account um and uh this more recent one right this is this is just this past year right um the the markovian modernism one yeah it's another it's an updated view on it's an updated take on on the consciousness problem on the one body problem um that question is also an author on um i think it's interesting it's an interesting um they're doing interesting stuff for the formalism with the information geology stuff um i used to be very um compelled by the the problem of consciousness but then my body problem i was very um i worked with that as my undergrad that was my uh so that was that was a big thing for me right now i was like interested in evolution and interested in consciousness a couple years away from the mind body problem before i come back to it so i'm trying to like i try to like keep my blinders on and focus on on like one or two problems at once yep philosophy has that feature of you pull on one question and then all of a sudden you're asking how do i know and who am i what are we it's because they're all very linked up and kind of like it could be the same with science you make a measurement about the gazelle and then you want to know about the cell but in philosophy it's

almost hitting you all at once and there's less clarity about how to structure those questions and that's actually part of what i think these conversations help clarify because in the example of the animal we know that there's a certain nesting or interactions between different types of biological entities and so we can coarse grain and that allows us to make useful decisions like about wildlife management or something like that again with an end in mind and a team and a method in mind we can do useful science and so the question is with the method in the team and the question how could we do useful philosophy and then maybe even move beyond these titles overall and just have the group working on the project with the approach so marco then mel i just briefly wanted to uh uh joining the social unconsciousness because i'm not out of it yet i'm still obsessed by it um so so so i think the beauty of microphone is or or dual aspect mode uh is is because the most beautiful thing for me for active inference is the emphasis and the primacy of mutualism right that nothing is without context nothing is seen in context um and that all their so-called identity so-called properties so-called t-loss these are just heuristics we've accumulated over the years in ignorance of the relational nature of these systems and so the beauty is that these relations of relations of relations these tangles of relations themselves become thing-like because they facilitate or allow for instantiation of a coherent activity i forgot what the papers i've been trying to find it's i privacy dig on twitter but there's a very nice paper about how some nerve populations that might for example hypothetically are about encoding cats this just put it very simply and they're just constantly chaotic until they get an input that is relevant or comes from sent data cats and then that chaotic high dimensional noisy stuff collapses into a particular order um and i think it's also interesting for you stephen because you also think about spaces and and this relates to a lot of well some people think it's too hyped but the work on manifolds and computational manifolds in neuroscience so the weird thing is you have this really uh you know a lot of neurons and population and they're related to a relatively challenging imperative problem but then for some reason when you analyze the data um you can obtain a manifold you just say well the neural population is basically just traveling on this low dimensional manifold so that's very the interpretation of that is another topic but but since you asked about the nationality i'll just add that yeah actually mel yeah it's what what you're saying just reminded me i had a thought yesterday i was i was trying to nail down one thing and of course you know there's this right you you pull at one thread and everything unravels um um i i see that there there can be a kind of systematic way of going but i i think philosophy is good it's not so chaotic it's it's right it's it's using the same kinds of of of conceptual tools just not with the formal rigor of mathematics but but there's

still kind of a uh system it's just so it's sort of like you know a mean field approximation or um solving a system of differential equations approximately right where you're you're um you're you're taking you've got you know these 11 moving pieces and you're taking 10 of them and your your either sampling over them to get an average value or you're you're approximating them and you're you're kind of like freezing you know you've got all these dynamic moving pieces but you're freezing the values you know so so to do this paper right um there's no one definition of what a model is right in philosophy of science there are you know 50 people working on models and they have you know at least eight different like distinct conceptions of what a model is and and what a model is rests on what similarity is what representation is what analogy is right and when it and then you you know you dig into these and they're all these different conceptions of what similarity is and what representation is and what analogy is um we don't know exactly what a theory is we don't know exactly what a uh the scientific method is right but to to drill down on one thing you've got to kind of stipulate you've got to kind of like average over the literature and find what it is that's that's a kind of commonality and what we think about a and kind of stipulate well you know no one no one agrees on exactly what a model is but a model is approximately this thing and in order to figure out you know what this one model is doing we can sort of stipulate roughly what a model is um and it it connects quite similar i think to what we're doing in math and physics but um just using language yes um mel i think you've taught me more in that like whatever three minutes than i learned from all my professors so far so thanks yay that's fun stuff i agree that was really awesome and again it's like philosophy and science we're almost converging on this approach if not just converging on the conversation because when you were talking about how many definitions of models there were i thought about how many definitions of foraging there were for insects or definitions that go beyond insects and then i thought about well when we measured it in the field we said it has to be walking in a straight line with mandibles empty between this time of day and this time of day and there's no single point that we would draw the line in the sand is it one meter is it three meters it's always going to be like you said it's about looking at the distribution of the literature if it's outside of the bounds of literature you're going to have a lot of justifying to do to the reviewers that doesn't mean it's wrong it just means that you're going to be having to make special claims like you're going to have to write an extra subsection in your paper to justify why you did that um but then if you use exactly what the literature did again whether it's right or wrong it doesn't ask to be justified and so people who are operating within that center of that mean field approximation they kind of just

point outside and they don't need to do an internal justification of their choices or of their approach and then you just said like you have to freeze other things so you did the scholarship to show there was this whole section on scientific modeling and then all you can really do is trace out the if then you said if you are feeling this way about models then you would be consistent in thinking this and it's not that it's better to be consistent versus have cognitive dissonance it's just that you can do the scholarship and trace out the if-thens of thought so it's just a fun way because there's probably people who are hopefully listening to this and curious about the conversation who are coming from the philosophical side or from a scientific or from an engineering background and so it's like there are similarities across all of our experiences like the need to iterate the need to freeze specific subparts of a model whether it's qualitative or quantitative and then work with teams and with corpuses that aren't just giving a single resounding answer for what is a model or what is a foraging trip so really just important nomadic ideas that we're broaching here so it's just good to take stock of that so marco and then we're gonna just walk through a few of these slides unmute marco um yeah i was saying you you've again relieved me of a lot of stuff i wanted to say um but but yeah i also echoed this um i think what we're seeing also in philosophy and science is moving away from or moving towards pluralism and i don't i don't think it's a problem that there's no consensus on the single model i mean the model is not a natural client it's a concept we we're using because our again non-natural or abstract processes of inquiry necessitated the phrase model and so a model itself is contextual or context dependent so it's kind of a fear relatedness or context lateness or normal bladeness i guess but i also like the the comments of now systematicity right so we've inevitably accumulated all these conceptual tools and inevitably people will try to find system testing between them and i'm kind of wondering uh in a weird reflective way if activision also can shed light or uh make interesting uh that phenomenon of of what we constantly said uh letting the fly out of the flyball we've surrounded ourselves with so many tools bonded together with systematicity without actual real justification uh and so we're kind of trapped in that fly bottle because we haven't had an alternative so you know who knows active instruments can fulfill we can stand stream so on the fly bottle um in undergrad i worked in a fruit fly genetics lab and the joke was you'd have to fly in the bottle and you'd say there's the fly in its natural environment because it couldn't live outside the lab it would get selected out so it's like no we're studying it in its natural environment now that was laboratory biologists who were just very aware of how selection works but it's such a deep insight into like yeah you are always studying it in its setting whatever it is and then you just broached

upon this contextual and relational thinking and then also that modeling it's not just about the object or the essence or the natural kind model is a verb so it's like the dynamical and the relational approach is coming and imbuing all of the things that we're working on so steven's working on performance mel is working on the history of thought and about the discourse that's happening at a very high level we're all working on different attributes here and then there's something that we can align around like a strange attractor itself and that's that we should probably be thinking about context relationships dynamicism multi-scale models so that's what crosses systems and that's really the fun conversation that gets to emerge in these moments good i'm glad that people got riled up by that so let's go stephen then dave then mel one thing that also comes to mind i think that's quite pertinent maybe in the broader context which everyone the universities you see it this interdisciplinarity transdisciplinarity participation um mostly it's words that the universities are getting into because they they start but that it raises massive questions because um how do you start to not just try and contest an idea within a field or you need to work between fields across fields you need to bring in context and multiple contexts and i think this you know question about how we actually start to understand knowing and at fuzzy levels of knowing which actually opens up potential to bring in people who are more marginalized because they have other ways of knowing and experiencing you know and they are better adapted probably to survive in some very difficult conditions compared to the the research themselves so sort of without making the you know the genetic comparison but you know the the whole ecosystem and bringing it all together so i i just wanted to bring up that question of like how all of this does relate to transdisciplinarity um and the need to do that with the world in the state that it is okay chill dave and then mel and then we're gonna move through sight yuck panks up and made a comment along the lines of your fly bottle um he found he could not work with white rats these specially bred rats when he's researching the evolution and the socialization of emotion because these are completely artificial creatures with no normal rat emotions you can't do those things to a a real rat that rat runners do with white rats what he took was your regular old brown alley rats the nasty horrible rats that everybody's afraid of raised them around people for a couple of generations was really careful not to select out anything and they just got to loving people that could play with you and come over to be tickled and they'd giggle and have a wonderful time and he says you know these are real things these are actual animals that could that survive in the wild i could let them go and they'd go out and live white rats would be just wiped out overnight okay fair to say that yes it's not about the exterior shades in any species mel on

anything else you want to say there i actually forget what what i was going to understand but yes chill it is very interesting and it's always the real environment and who's adaptive and who's not it's it's not something that can be speculated before the ecosystem emerges because most species go extinct and there's insects that had a strategy for 100 million years and then they went extinct and it was always situational and interesting questions though so we're gonna pass over the abstract and also there's about half an hour left in case people want to put any questions in the chat so in the roadmap i think it's a good place to pause on our road trip analogy we can start a we can see some of these terms that we've brought up in the discussion reflected by sections in the paper so um and by the way if anyone has to leave that's always chill um mel go ahead i have to leave yes thank you i saw i saw the time drawing near mel thanks so much for participating in 14.1 and two it was just obviously great discussions and you know apologies for bashing you after you leave no thanks for coming on oh please do yeah in any other future discussion you're always welcome to contribute your perspective so we'll hope to see you soon fantastic thank you cool so nice we can all you know continue to go around but really interesting stuff we did hear a lot of these things let's look at this table and maybe are there any terms that we haven't really brought up we've talked for example i'll mention mean field i can see a few things already but maybe someone can raise their hand and pick a term that they don't see how it's fitting into the things we've already brought up because these discussions it's like it's not the end-all be-all it's just a couple of hours mel probably spent hundreds of hours writing the paper so we're just sharing our perspective seeing at the first pass how these ideas are associated and then there's a paper and a literature and a research community to delve into if some connection is interesting so someone's like well with the connection between 6.1 and 4.3 was really exciting so dave what would be something cool to go into yeah um i first encountered the term free energy in economics where it means something just dramatically different it's a good thing in economics um but the closest that i've seen to a use with the connotation that that we've been using is as anxiety free floating anxiety paralyzing terror then i don't know what's wrong is that uh off the wall this is fun um it's definitely an unconventional economics interpretation of free energy but it just shows how when using terms it's always important that we understand where we're coming from because in a way that's what this paper and also several other people papers have um brushed upon which is there's a thermodynamic aspect to free energy gibbs free energy which is like after you ignore the activation energy so whether or not the kinetics of the reaction are such that it does go down after the reaction has occurred ΔG change in free energy is like how

exothermic the reaction was is what draws the reaction forward and so you have a candle it's like water in a dam it isn't spilling over and then the ΔG is negative so it wants to burn and then when the activation energy occurs it does burn so that's the thermodynamics free energy there's more to say there and that is actually addressed in mel's section on statistical and thermodynamics the origins of this free energy term in this setting are from again statistical and but as that became more and more integrated with information theory what's called the epistemic turn there became more and more of a marriage between physical interpretations of free and more informational or statistical and even bayesian and computational understandings of free energy so then we were at this point where free energy was being used pretty fluidly as a metaphor in machine learning by analogy to those chemical reactions but now in the statistical setting and that set the stage for the innovations that fristen specifically made which we can go into but that i hope was a little recap it sounds like economy drew this free energy term and then just to close on that you said a generalized anxiety yes it's the differential between um you know being too anxious and having too much surprise and being too um rigid and fragile though converging to a perfect optimistic model is not a bad thing so it's sort of a gradient descent on expected free energy because just the reality of the situation is that we're not going to come to that perfect model really in any meaningful respect it's just not how but these are there's other you know little pitfalls and land mines and things and these are all papers that are in a network like the paper that we're going to be reading in 15 about is the fep realist or instrumentalist is it actually how things are or is it just a scientific instrument so those debates are being skipped over in this section but that's a little bit how free energy got from statistics um and statistical thermo to information theory thanks um uh just fair warning i'm gonna be a bit expected here but but um i always like when people uh make nice metaphorical connections so so they've um yeah i i i would want to um um build upon your your intuition so so um like mentioned we have thermodynamic free energy and no matter what these thermodynamic free energy can be used for work right but then we can ask okay is the work that's going to be done is it going to be useful or not for the agent and so this ongoing process now we're just talking about thermodynamic energy this ongoing process of basically channeling or or allocating or distributing the thermodynamic free in a way maximally useful for the agents is a challenge so for example it can try to challenge somewhere which will evoke a certain inferential challenge but when it fails you'll get negative effect right so so so one part of you sends a challenge the other part can't resolve it and then you feel it so so yeah i i i kind of feel like it's not that wrong in my opinion to to see that because anxiety is very uh

specific it's it's basically kind of like fear but it has no targets right it's generalized so you're afraid of spiders but you have social anxiety because the whole domain of social interaction is just fraught with uncertainty but without the availability of a specific reason why uh your you're negatively uncertain or in an uncertain negative way that makes sense marco i'm going to build on that because it was really helpful so you mentioned how the fep has to be deflated because it's like generalized bad it's not the same is bad but it's something that is being optimized whether you think of it as minimizing or maximizing depending on whether there's a negative sign so it's the kind of thing that we're working on that's the key point and it's inference what we saw here is that we're going to draw a physical analogy from the way that chemical reactions indubitably go downhill the candle doesn't unburn so we want to design governance structures information schemes economies that are like a candle and they just irreversibly according to our experience they just play out you don't have to specify the micro level detail just as an analogy so we want something that's going to go quote downhill in governance space the thing is we don't just want to simply do inference on it so in the machine learning area it'd be like okay what's the data set let's fit the estimator let's reduce free energy so that we can fit the estimator on the descriptive like on a big k-means model how many clusters are there then that will be an inference model we're actually doing free energy gradient descent on what policy selection that's where control theory and cybernetics comes in because we're not just doing descriptive analysis of parameters of large data sets we're actually doing prospective or anticipatory policy selection in light of a deep knowledge reflected by embodied priors in a niche so that is the oriented turn where we go from again just pure free energy minimization to do inference via tricks and heuristics like variational base and fristen takes it to the next level by doing a few things not the least of which are incorporating action conceptually doing inference on policy selection which we've seen embodied in the models where like the π is influencing the b so the way that the state estimate is changing through time the way that s and the b matrix are related is being modified by policy so policy is what free energy is plugging into so we're doing free energy descent on not just descriptive attributes of the world or any smaller large data set but we're doing a prospective free energy grading descent on policy because that's about our world model and how the states the world change for some people i know i was a little repetitive but that's really some of the key history and that helps understand how statistics and thermodynamics and bayesian inference which already is a big synthesis to make how all those areas in information theory and computation and everything that's kind of quantitative that we're talking about how did that get involved

with ecological psychology control theory cybernetics and then even into this deeper relational way of so that's a little bit of that and it's an intersection that will you know embroider over many times but it's really a great story and it's awesome that mel wrote it up in a way that isn't just backroom lore but it's something we can be like yeah it's in 3.1.1 in mel's paper so that's what's great about literature stephen yeah thanks daniel just got to take a breath because you that was really uh that was quite a lot of useful information i agree with what you're saying there and i i suppose one thing that is good to bring it back to that gibbs free energy is you know you know there are times when for instance ice when ice freezes and the um the actual um free energy um is such that it gives it acts as a force it acts as a force so that um the the the the lower energy molecules um are not necessarily purely favored and it takes a slightly different structure and you you know you end up forming ice of which um you know it goes to a state which is um not necessarily energetically favorable you know so but that has that that dynamic is still something that can happen over time with equilibrium so you've got this ability as you mentioned to to sort of um to select policies um outside of equilibrium to to um to to keep going and to work with that um but i think that what the free energy does and what this talks about that's a bit different to how a lot of people talk about and stuff is they normally work with these phase shifts and this kind of like if you put enough molecules in a box and get them to be constrained in this way and that it will emerge that you'll get the ice or emerge that the ice becomes water um and that seems to be like the prevalent way of thinking about emergence sometimes in kind of economics and some other cases it's just the interactions under the the the temperature and pressure but what this gives away to think about how it can be um innovated upon and thought about in terms of um policy selections happening at multiple scales and uh the the challenge is it depends which scale you go at and you're talking to try and explain it to someone you know so the cell versus the organ versus the organism but i think that um that that's something that this um really helps to unpack um because you can't deny that this is not just emergence as you put everything together and they'll interact and it is it involves some sort of um sentient uh or some sort of way of building up but starting from a non-intentional base so you and it's got both which i think is really cool thanks steven i want to mention two points so you talked about a phase change and so that's a reversible phase change with like melting and thawing so it's a little bit different than the candle you know melt thaw resolve itself into a dew that kind of stuff so slightly different with a chemical reaction versus freezing just from the chemical side but you mentioned it was not favorable for the water molecule to freeze from the liquid into the solid face so go from freeze from liquid

water into solid water but actually it is it's that as the thermal energy reduces all of a sudden it clicks into a new differentially configured arrangement and so that's what the phase is is like when you're at absolute zero i assume most substances are solid like they're in whatever their ground state configuration is which is like a close packing it's a crystal of some kind it's a solid and then there's some temperature t where that crystal melts where all of a sudden it's like the iron is vibrating it's a crystal it's a crystal it's a thousand degrees it's 2000 degrees i don't know what the melting temperature is but at some point the iron is breaking out of its crystal and goes into the liquid phase and then there's still the inter-attraction of the different molecules in the liquid even the iron and then at some point it vaporizes when you're even hotter than that and plasma and so on so these phase changes are actually they're about the temperature and then if you want to go to that fallacy you're kind of mentioning it's like people think that you're gonna get a phase change by adding more into a box or by stabilizing it but it's like no if you're at the wrong temperature you're never gonna get ice you need to do an enormous amount of pressure to get solid above zero you have to crush with like millions of pounds of pressure but you can make water into ice at like five degrees celsius you just have to put a massive amount of pressure and so the question is how do you do it with zero pressure that's by changing the temperature and that's where this multi-scale yvonne in the chat 2282 fahrenheit for iron thanks for that super clutch then the second point just quickly and anyone else then sarah is which scale and you said the cell or the person in the society and to me that's making me think of helen longino's book on pluralism because she discusses how the different fields like the sociology the geneticists the molecular they have stories about aggression and sexuality in the case of her book and depending on which method you're familiar with which affordances you have at hand which aspects of the world you're sensitized to you're going to tell a different story a molecular biologist is going to have a protein binding answer to for example a pandemic a sociologist is going to know about some social intervention and so there's all these different people and it's not just that one person's worldview is like the right model and someone else's is the right one that's kind of this question is which scale it's not for any one person per se to say but those are the questions that we need to gravitate around to make action that isn't unduly biased towards the molecular towards the social etc but rather something that works across those levels so really interesting stuff steven thanks for like always highlighting how it comes back to bigger questions like that sarah and then marco yeah i mean maybe just to throw some more monkey wrench into this um something about what you said really really um reminded me of um the phenomena of hysteresis and i never really

thought about hysteresis this way before but i was like oh jesus i mean it is true i think that hysteresis is an expression of of hysteresis historicity um you know of history and and so um you know with with fep i'm not sure with active inference maybe but like i often ask myself how these various you know like you have an entity as you know by way of your model or natural kinds or whatever it is but you know when you hook in these enti you're kind of you know weaving or whatever these entities together um by way of something and and hysteresis is always been interesting to me because it's like that extra bit of dimensionality that phase if you want to call it um that can hook into something one of its neighbors and so i often think about the um like let's say with fep or active inference you know like if there's only like one interface that can be used to hook to a neighbor you know maybe it's phase change or whatever that that moves out to an adjacency um so like how many different degrees can be used to connect as an interface to various neighboring agents type thinking and i know that's left field i don't but daniel if anybody can take it somewhere you can or marco i don't know marco definitely take the first swing at it or anything else you want to talk about thanks i'll do my best um so let me just grab my thoughts before i lose oh yeah yeah uh sarah but what you just said i'm not sure if you find it relevant but one of my favorite papers is from um avena konigsberger and all of spawns and other people about routing strategy so so i i personally am very fond of the conceptual set of flows and paths and routes right because you know quite selection can also be seen as navigating a certain policy space so this paper spectrum of writing strategies effectively is tackling an interesting question of when you have a given system how much non-local how much uh information in its adjacency its context doesn't need to incorporate right so as it expands let's say its domain of discourse or its inferential context as it expands there will be a non-linear growth in let's say the dimensionality or the rank with the involved elements as it were and that's a cost at the same time if you don't expand maybe you're just not able to resolve a certain differential challenge and so this trade-off of of how local and how global or how expansive you want to go for a certain differential context um is i think related what you're saying right so how many degrees can you afford uh when it comes to uh selecting how you go about selecting a policy um if i answered your question i'm not sure it's right but i linked it in the um youtube comments thank you um and the other thing what we're talking before is thermodynamics oh yeah i am still not mathematically um competent enough but for those who are i linked also uh in the youtube chats to um a article by uh john carlos baze who is a huge figure in mathematics moments and he wrote something about entropic forces and when i discovered this it was just mind-blowing and amazing

because basically what he's doing is he's licensing talking about forces as governed by entropy so so it's basically about whether the temperature is static or whether it is either thermostatic or or motion centered was it anyways um but when the thermal the temperature is static then the forces are better described in terms of entropy which i wanted to emphasize because we had this discussion a bit with uh steven's point earlier and i i often like saying to people when they don't normative principles there's always at the very least the ultimate normative which is the principle of least action you do physics the cycle of thermodynamics there's always that no matter what every system has to be governed by that so given that we won't deviate from that law we can say the brain is governed by a principle of this action but clearly it also is governed by uh imperative to systematicity uh or mutual inference this constant co-regulation and so the question that i wanted to explore is kind of how does that link together what how i like to what if you phrase it is basically these kinds of systems are effectively cultivating nature to do the work for so how do you transform the dynamics governed by the uh by the ultimate principle of fiscal peace action how do you nudge that into also being systematically conforming not just physics but also your inferential imperatives if that so this is a very subtle and really interesting point marco thanks a lot let me try to trace a little of that so you were pointing out how bases work on entropy and some other aspects of information theory and the fundamentals of physics it's almost like integrating statics and dynamics into one more unified field and you or or you're muted but is that correct or where where did that and i i i'm trying to trace how this comes into the thermodynamics side continue though i'd like to understand that a little better yeah i mean honestly i think we should have discussion maybe uh yeah later because it's actually brief post i'm not sure if he has a formal paper about it okay um but in general base has a lot of interesting work about entry and biological system again you know way out of the element there to be for me but i would really be curious to see what what more formally uh competent people think about these terms but the thing about entry force is just a very simple blog post and it's it's it's it's not that complicated okay we'll read it discuss it and maybe invite him or anyone else to be on a conversation so we have about you know 10 20 a little bit of time left so let's think about some thoughts questions we're taking forward with us but then we can kind of flip through a few more ideas sarah to your idea of hysteresis and historiosity which are both two kind of similar sounding words but hysteresis is just the system where like when something happens it doesn't return to exactly the same state and historiosity is just the idea that things have a developmental history like even if you reelect the same person they're older it's a different world etc so then you also mentioned the connectivity

of agents and then that made me think about social agents and how the social networks have a hysteresis element to them like if there's a fracture it's not like you just zip it back up like a sweater it's like gonna be something that has to be um recovered if it's recovered exactly or not exactly it has to be recovered in a different way so i kind of see where you're coming from with the hysteresis and the connectedness of agents now where does active inference play into this if it is related i think it i think there's some elements because we're using a lot of these words so it's kind of like these networks of thinking about these topics i'm not sure where one would take that going from the idea that it's the networks and the relational insight that defined function of the system and then hysteresis and thermo i'm thinking of a few things though marco yeah well i mean i would love to hear your thoughts about it so i think i there's a bit of a tension because often the formless is described in relation to mark of change which is explicitly about just being dependent on only the last state right but i i i think what sarah was kind of alluding to quite rightly so is is when you have these uh not just one system but your interconnected um then the a a past it of a particular system might come back with a detour support and and i think that's kind of what she's pointing at even though you could of course define the mark of chain for every point etc um on a different level you can say the past states for example that system expressed an influence into its larger context and then later another system that received that influence uh responded with again a return signal to that initial system that makes sense yeah this is actually what i'm always obsessing about um is and you can analogize it to complex numbers as well you know like the hysteresis adds that it pops out an extra dimension that something else can hook on to um and that's what i'm i'm i'm always thinking about with respect to you know um i don't know if not even well-versed enough to even say fep but what i understand about active inference you know like that the model seems to just creep in a forward direction but there's no side chains like there's no there's no side hooks um and so i'm always i'm always like yeah but how does that how do you do i don't know like i'm it's and it may just be my limited understanding but but i come from a place of thinking a lot about hysteresis and memorisers and stuff in it and i can't quite get those the math to answer my question did she elaborate on side hooks i think that would help agree what would a side hook look like not a sky hook for those dennit fans out there well what i mean by like let's just say you know like you have real numbers and then you have complex numbers and i think of the complex plane as like a side hook um where you you add an extra dimension basically where other things can interact with that number space by way of the the complex side hook but it might be a too limited paradigm for me like i'm throwing out these random

ideas that may or may not um work together but that i don't know it's let's say uh this is like totally sci-fi um you know like you have memorisers and they're in that but they're going in like voltage amplitude one direction but then they have different paths um as they go forward versus reversed and and so that that phase that kind of phase relationship um might offer a relationship with something else that's related to phase or related to um you know whatever the physical dynamics might be around that so i don't know like when i think about um active inference you know like they they're just like well what's the happened in the past and what's going in the future what are my prayers but there's nothing on the side that um sorry it's as good as i can do i don't know if that's helpful you know you rile you riled everyone else else up so it's great so we're gonna just slowly we don't need to end exactly on the hour if anyone is gonna um leave on the hour that's fine just um maybe speak your piece as your closing piece but we'll stay on a few minutes after because yes this is a really nice topic so marco dave stephen yeah thanks i think i understand what you mean with side hoods and i think maybe what you're what you're um bumping against is is what i also find uh problematic so it's good that you raise this so so the the the the equations the formalism for free energy and uh for fep and active infants they're standalone they're like the most naive equations you can have right but the whole point is that when we apply then we have to fill in the things that were previously naive we have to for example only now only very recently we're going to multi-agent systems with active influence but that was always implicit in the research as in from the first paper in 2009 2010 it was already implicit because it explicitly talked about how it would be multi-scale how it would be um multi like nested right it's all these different systems all doing frenzy minimization in relation to each and so that that codependency right these mutualistic relations those are the scientifics i believe so for example uh if i'm not abusing maths here you know the decide for complex numbers is it's imaginary until you square it right so it's a potentiality until a certain operation occurs so in the same way when a particular system expresses an influence into its context it isn't affecting that system anymore until something happens when it returns right so so the fact that it's always situated in an extended environment um that a particular system is very naive about actually that it can't know everything exactly that it has to kind of trust or upload certain um influences to its uh context is kind of that side i would say um does that make sense interesting i think that once i look at the math i'll keep that in mind and it might start to it's that's really helpful don't don't look at the math don't look that's the point right you'll hear v maps describe a general uh general principle for how they are by what they are governed but they are governed in terms of states

and relations to other systems right so i i kind of confused because you've alluded to it many times before how they're all connected to each other right so i think that's not in the math oh wait i'm just going to continue with it i think i think it's a little bit in the math though i think if i keep your this is how i think or how i learn is i keep these kinds of phrases and statements in mind and i look for them in the math and then it becomes clearer so i i think yeah and you know and you've talked also your metaphors are great you know like you've talked also about context and shifts in context and and you didn't say this but my brain did this like collapsing or like something becomes um so i'm gonna it's really helpful to keep these things in mind as as i'm battling my way through you know these formalisms um so yeah i'm just gonna ask daniel because danny was way smarter than me i know there's a mathematical relevance here because of matrices you have a marginal the diagonal and then the other entries are for the conditional conditioned situations and that also reminded me of the scientists if that makes sense oh yeah we're gonna we're gonna return to side hooks but let's go with dave then stephen okay um hysteresis and side hooks uh the first the most elaborate discussion i've gone through on hysteresis is in renee tom and chris ziemann on morphogenesis structural stability and catastrophes in there um the hysteresis are refer to hidden states or relatively inaccessible states so these these amazing things like the the financial collapse in 2007 and 2008 where you just absolutely fell off a cliff and none of the models said that would be possible or in uh say a psychotic a psychotic break the most stable guy in the community just goes totally wild or good things too uh you know a religious conversion that turns the slaver into the guy who wrote amazing grace well now more recently um gerald edelman emphasizes relatively inaccessible states a tremendous amount both in his work on immunology and in neuroscience the term he uses degeneracy which he borrows i believe from analysis numerical analysis to mean uh something roughly like a function whose inverse is not a function of multiple ways of coming up with an activity which from the outside looks exactly like another activity but it's implemented in a completely different way and then in a professor fristen's paper uh that was uh suggested reading a couple of weeks ago uh on where he's talking about uh as uh ascent uh climbing gradient um he says it seems as though what's going on inside the markovian blanket is it's learning about its environment but it can't observe the environment how can it be learning about something that it can never observe so i don't know where to go with that but it all sounds like it kind of would fit together and make a nice paper thank you dave for mentioning that and lots of great points stephen and then i'll also go with the side hook and histories this discussion yeah i think this one thing this discussion really talks about is the the

relativity between what the organism in the markov blanket the generative model and the niche and the environment so the world out there can follow all the rules physics and it could have a lot of side branches and it can go and behave in all these sorts of ways and which we don't know about some we can track through the entropy some we will know because the niche will hold it i my cup will still be on the table so some of the the kind of the stuff about least action has a benefit of maintaining a trace more clearly in the environment so that there's an inch i think this is where it's quite interesting is how much the the the way of coming in it from both directions i'm starting to see what karl frisco meant there in some ways is sometimes it's about the model of the outside and sometimes it's what's out there it's been shaped and humans we're doing a lot of that so when we shape something out there what what is available because they talk about the adjacent possible a lot when they're doing work with people so is it is it that maybe you know as an animal you know you can infer much more diffuse abductive kind of things if it's within your sensory range however it needs to be within the adjacent possible whereas in the niche you can do things at distance you know um you can fire something over the hill and then you go around next week and you see where it landed you know um which is kind of so anyway i thought that's quite interesting i think that um really does speak to this two-way movement and the fact that some pieces are more about you know it changes the name of the game for emergence for sure beyond pure physics into this non-equilibrium and i'll put a paper because i found a paper that talked about non-equilibrium chemistry when i was looking and it was interesting it said that there's almost been no non-equilibrium chemistry until like they only started doing it in the 80s and 90s and it's like then you do it's all this at equilibrium type work where you go from one to the other and then of course all life is a non-equilibrium so it's kind of um interesting because you've still got both going on it's just happening in the in the niche stephen excellent point let me go on just about in reverse order there's a few excellent things you said long-range action through niche modification that's stigmergy the ant can't do long-range instant communication but it can instantly put its abdomen on the ground three months later another nest could instantly detect it so it's at an instant with a behavioral focus but it's actually long range with respect to action of that one nest mate on another so then there was a discussion by dave about how fristen was mentioning how how does the organism learn or how does the system learn when it can't directly infer directly you know measure the states and that's everything we've been talking about with the observables the organism is actually observe getting the observables and using those to do state estimation on the hidden states of and then it turns out that the observables

don't have enough instantaneous information to reliably identify what you care about which is the latent causes of the world so you have this imaginative generative which includes that space of the adjacency you know the adjacent possible that's weighted by how likely it is and a bunch of other stuff that makes it more like a candle burning than just a marble on a hill so it's a little bit more nuanced because we're talking about something that's like a bayesian computation at least as the scientist uses it and again that distinction between is quote what's happening realism versus is this just a instrument that scientists are using in the world that's going to come up next week in 15 and so that's like really interesting all right so the side hook i thought if there's a idea with many sides then maybe it has many side hooks because like it's clearly by analogy to the complex number so i like this idea though of jumping out a and that seemed like it was related to what stephen said with the adjacent possible and with policy imagine imagination it's like if there's a wall in front of you and then you're only on the number line you're gonna hit into the wall but then if you are logistically in a different dimension then you can go around it or you can figure out whatever policy it is so policy and imagination is like popping out to a different dimension where then you can enact a policy but if your imagination isn't connected to the manifold of reality your action selection would it's like a fantasy it's like if i could fly i could just leave this building it's like yeah but the e matrix is not that way so it can't happen so it has to be imagination with the affordances and then also just one last point on the statistics so the markov chain that marco um i believe mentioned so like a markov chain with a memory of one would mean a statistical process that's only dependent it only has a memory of one whatever the previous state was that is what influences it and then it's almost like as a statistic statistical model so again whether we think of this as a scientist instrumentally with a model or the organism actually embodying this model let's just say that most things have an influence of one time scale but then once in a hundred times it has like a range of a thousand so there's two models you could make you could have a model with a markov chain of a thousand so the different action sequences of like every single possible last thousand days and that model is very computationally complicated but if you've had enough observations it's unrealistic you'd be able to fit that like to a t versus the more simple model which is just give me the memory of one so i can have just one bit of memory and then i'll be surprised when the other thing happens and so that is the compromise of the modeler of model accuracy minus model complexity do you make this markov blanket like simple like a heuristic or a bias and then be surprised by particulars or do you try to fit this ridiculous particularized model for every single case but then that loses all efficacy because

then you're tied up so specifically with the very specifics of the model that you learned that's like overfitting so i know that's a ton of points but those were all like really awesome comments and i think we're getting to a lot of the stops on this little road trip but let's just head to the last section but anyone if you have any final thoughts or questions that was yeah basically the only other slide was this previous one with a little bit of and then just the usual questions but i'll just say thanks for participating and stay in communication with everyone and then let's pull back to uh the road map or to here and then each of the four of you thanks so much for these like just amazing conversations so each of you if you would like to just take a you know we'll go one last round of people want to so whoever wants to raise their hand otherwise we can go to this one i guess so this slide was kind of starting with the physics on the bottom with a focker plank and a flow so that was what dave brought up and the way that this conceptualizes a flow like on a hill you can have like two orthogonal curls two orthogonal factors one is kind of curling around the hill at the same elevation that's like what you see on a camping map like the iso contours and then the other one is like a ruler right on the hill like straight up and down and so not to go too much into physics but the ruler is like the Δg it's like the gradient descent is the candle burning or is are we in a flat spot so that's the ruler is like are you skiing fast downhill or are you not skiing fast downhill and then the iso contour is like is a steady state that's like when you're um when you're doing a chemical reaction there's always a forward and a backwards reaction now the K_{eq} the equilibrium coefficient for the reaction's completion might be 99.999 percent like a candle but actually the completion is really high because it's so exothermic but stephen from chemical manufacturing you know that reactions that are barely favorable you're going to end up with like 52 percent of the product that you want and 48 that you don't want or something like so those are that's that decomposition right here with a helmholtz decomposition that there's the solenoidal curl and irrotational or divergence components so that's kind of a thermo info connection so when people are talking about gradient descent in machine learning they're talking about like just like chemicals are allegedly physically skiing up and down these slopes also in machine learning you're doing informational skiing on these slopes and then that's where you get these bayesian statistics and thermodynamic ideas the key insight in the the integration with control theory happens here basically if you're going to be maintaining that far from equilibrium better than all the other models like you in the niche then you need to have a generative model not just of state estimation because that's sort of like the low hanging not really fruit even if it is low hanging or not it's just not really fruit because action is what the agent needs to do to stay

far from equilibrium whereas a machine learning model on our computer it doesn't need to behave well to survive because it's like under not really the kind of selection pressures that we are but we need the action element so then this imperative to be self-evidencing as an information forager means that you get all of these cybernetic characteristics and that you see systems that act as if they're doing the optimal experimentation given their priors reflected in the generative model in their niche so that's where you get this sort of like forager information optimal information gain principle that's what's needed to keep the action selection effective because they're far from equilibrium steady state that has to be subsiding despite these things that's what gets subsumed in free energy principle as discussed in mel's paper and then on the top are corollary process theories so it's like okay whoa i guess that would have to be true for any candles i saw out there how did that energy get so organized so it had to be an active candle or it has to be part of the extended niche of us right there's no candles that's like the watchmaker analogy and then how does that happen and these are falsifiable process theories because theory falsifiable empirical unique explanatory explanations and predictions um this is where we see the falsification happening and so we discussed in this paper like some of the already kind of i guess less in favor process theories like the gibbs filtering or gibbs sampling which is done on computers but is not apparently tractable on biological architectures in the way that these um more established or i don't know uh productive not falsified yet there's some good critiques of bayesian brain in the paper in 15 as well as in um the 2018 interview in alias with fristen but um yeah and then what else because the whole idea is that like as free energy principle gets shored up and connected laterally to a bunch of other things there's going to be other process theories active inference is about some kinds of systems but who knows there's other ways we could think about it and so i don't know what does anyone think about that other than that we'll head into our final section marco yeah um i guess i can add a hopefully pleasant perturbation so one interesting um thing to to raise here for this slide is um it's conceivable that active inference might actually be more solid in its conceptual framework or structure than actually the formalism for example you could maybe for example not use kl divergence as a distance metric but other distance metrics um because the kl divergence is i think also in the self idealization right in the end you're going to have very a lot of messy stuff in in real systems such that even if you're going to model certain belief distributions in the action implementation it's going to be extremely unlikely that it perfectly follows this um a particular conception of distance to be minimized um and so i guess i'm just urging for this expression of what can come out for me that that that um oh okay i personally just like to

encourage people to do to be playfully exploring different variants of the free energy principle because personally i believe conceptual uh model uh taking it as a conceptual model is better than as purely a formal but that's personal yes well thanks for expressing your thoughts as an action suggestion stephen and then sarah yeah thanks for that description daniel actually i'm going to go back and be able to use that as a really nice reference point for lots of information i like this top piece and i think yeah the bayesian brain was there as this kind of integrated idea and then they thought well let's have this predictive processing process and um i think marco was saying like some of the assumptions then for it to go to you know we the math is there to make it solid but we you know we can think beyond that because then they say active inference is really an active inference which goes another level and the math that's not a math thing that's like okay so if it's n active inference and then does there need to be like you say what else is there i'm actually quite interested in potentially been looking at some ideas of some spatial layers like we talked to had that thing about projective consciousness and some other layers like what are the what's the layers beyond inactive that i need to be there for these processes to be plausible and who knows that that would be something that we'll talk about in later date but i like this this slide could actually be something that could be played with in the future as a learning tool because i think it's quite helpful great thanks so much steven sarah oh the trail might be kind of cold now but you super had my attention um on on yours you know you had your arrow moving around and you said you know this is the place where there's like a connection between thermal uh energy and free energy and i was like where because i mean you know math wise because i've that's always just been this super separate world that's annoying and i can't really put it together so i want to find where that is and then go find it in the math you know that the word description in this really nice slide and then the math and try to figure that out great call great call um i agree it's why one of our projects just to sort of like talk a little bit about the lab since everybody here is basically a participant in it it's one of the reasons why having the ontology and the structure of knowledge for the equations is really important because we want to be able to say this equation in this paper is where the variables in the you know f from this paper in this other math area actually we're going to call it this letter and here's this and that's where there's a point of contact with this and then is this one following from that one is that one following from this one those are the kinds of things that truly in the frontier of what we can and then understanding how to bring directly from these analytical mathematical phrasings to the control theoretic in the policy selection that's a non-trivial it's not a velcro interface that's a serious maneuver that involves also moving through as

pointed out equilibrium and steady state and far from equilibrium non-equilibrium all these different things in this metaphor or is it not realm of info thermo so lots of exciting work to be done and traced good points and then also yes steven just you pointed out it's um active inference imagine if someone said in 2010 well active inference is simply one time step and it's xyz it's like it was and then now it's something else we've talked about and activism we've talked about affective inference now is the variable in affective equivalent to answering philosophical questions about affect no but neither was the variable affect in any other modeling framework which again makes us pull back and just look at that instrumentalism versus realism debate so just the realism would be this is actually telling us how the world is arranged and what it's doing versus just this is a framework that is being just like mel's paper this is one of the main points is with respect to science it's in the realm of science but what is it in the realm of science well it's not one of these falsifiable theories up here it's something that's more like natural selection at that type of tier but it's like a theory generator and and that's why if we look at the road map there's um 14 sections on kinds of models because this is like the system that this is being slotted mel could have or anyone potentially could go the other and say no it's not part of science it's part it's okay so you can debate about where it is with respect to the demarcation problem but since this is the paper we're in the journal club on we're saying given that that's the view being defended the literature has to be integrated in is not theology it's not poetry it's actually scientific modeling philosophy of science literature so then that will help us connect it to like okay but what does the variable in a linear regression mean is that realist versus instrumentalist interpretation nobody would say it's realist it's not there's not like a line in nature it's making one you know group of plants a different shade than another group of plants so that's sort of the process that this paper is engaged in is interfacing better so that we can understand what kind of thing the fep is and uh there's a few uh exhortations like this is it true what is it true of how do we empirically know that it's true and mel argues that that's a category mistake so it's like saying any other category mistake now think about these questions in relationship to the fep being like that so again great paper we'll just go to the closing now but just thanks everyone for being a part of 14.2 and all the other sections because this was really like two great discussions i'm looking forward to that people can raise their hand if they want to make final comments but i'm really looking forward to the with the authors um hippolyto and van s on 15 for both weeks and that will be interesting and we'll confront this realism instrumentalism discussion and then also um the zero videos will be a dialogue so hopefully more and more people will get involved with making the summaries

of the videos and introducing them from different perspectives different lengths and each paper is unique instead of authors and set of people who are approaching it so i just it's been really awesome to see people engaging with some of these truly long-standing and fundamental in a new light and then also just co-creating with first just participation in live streams maybe even just by watching but then anybody who's watching live earned replay is always welcome to join or find out what we're doing in the lab so it's just been a really good you know we grew a lot during 14 and so is good times so any other final comments that the panel wants to make yeah i'm gonna i'm gonna grow some neurons now this was awesome peace marco go for it yeah i just wanted to um ever since i first joined i can't find the survey you keep mentioning i have no idea where the survey is so maybe get to tell us and the viewers where to provide feedback sure well the survey for live participants is in the event calendar so it's in the calendar invite and so yeah for those who are listening to the very bitter end the drags of the active basically the single source of truth is the calendar invite it has the slides that are accommodable the paper the video url and the feedback form and then the feedback form you know it'd be awesome if everybody fills it out but then the feedback form for everyone else is just email comment threads youtube comments leave feedback publicly or privately we'll always respect both public and private feedback and let you know how it influenced our trajectory so but for live participants we have a special structured form that we can always modify and evolve as well so that's a little insider baseball but always happy to talk about that because what it is we do is it not steven and sarah on the discord and all the decision making maybe it is maybe decent but it's been fun so no it's great a very good explanation and uh i enjoy it today right like so i've got to grow i think i'll grow a spare brain and then i'll fill it up and put it back in my head just force it off source it put it in the papes all right good to uh talk to you all i'm gonna terminate the live stream